

App Dev 2 Project Report

1. Student Details

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About Me: I am a Student at IIT Madras BS Degree program with a deep interest in web application development, data-driven technologies and AI related applications. I enjoy building meaningful applications that combine learning, analytics, and user experience.

2. Project Details

Project Title: Hospital Management System - V2

Problem Statement:

Hospital Management System (HMS) web application that allows Admins, Doctors, and Patients to interact with the system based on their roles.

Approach:

The app was built using Flask as the backend framework with a modular structure. It allows patients to book an appointment for the registered doctors in the respective department with ease. The admin is pre-defined in this app who can manage the doctors and patients and also keep track of the upcoming and past appointments.

3. AI/LLM Declaration

No use of AI/LLM.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Vue2 JS (CDN)	Designed Frontend pages using Vue2 JS framework
Bootstrap 5	Frontend styling and responsive design
SQLite	Lightweight local database for storing user data
RESTful API	Backend routes, controllers and logic design
Redis	Frontend caching
Celery	Perform Backend jobs

5. Database Schema / ER Diagram

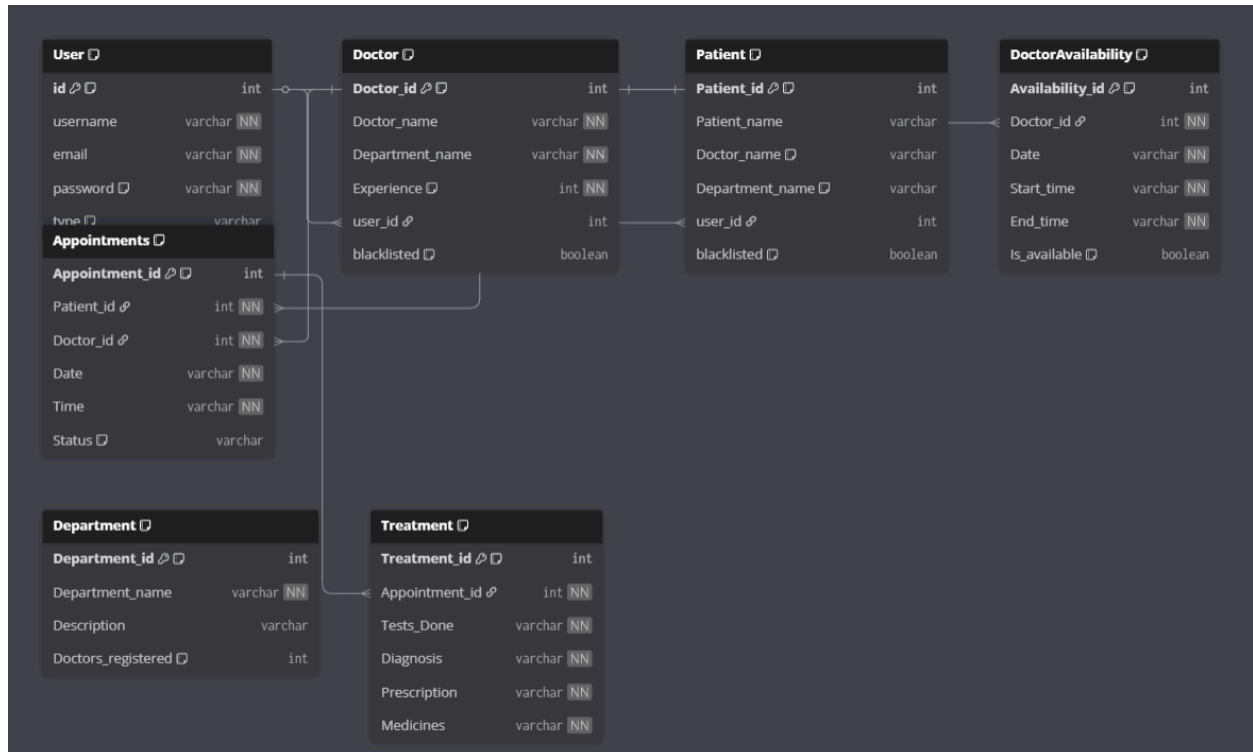
Tables:

1. **User** — Stores user profile details (id, username, email, password, type)
2. **Appointments** — Stores the appointments of the patients and doctors (Appointment_id, Patient_id, Doctor_id, Date, Time, Status)
3. **Treatment** — Stores the treatment history of a patient after an appointment (Treatment_id, Appointment_id, Tests_Done, Diagnosis, Prescription, Medicines)
4. **Department** — Stores the department names and doctors registered in that department (Department_id, Department_name, Description, Doctors_registered)
5. **Doctor** — Stores the information of a doctor (Doctor_id, Doctor_name, Department_name, Experience, blacklisted)
6. **Patient** — Stores the information of a patient (Patient_id, Patient_name, Doctor_name, Department_name, blacklisted)
7. **DoctorAvailability** — Stores the appointment details of a doctor (Availability_id, Doctor_id, Date, Start_time, End_time, Is_available)

Relationships:

- One-to-many: Doctor – DoctorAvailability
- One-to-many: Doctor – Appointments
- One-to-many: Patient – Appointments

ER Diagram:



8. Video Presentation

Drive Link:

 App Dev 2 Project.mp4
